

Unit 1: Introduction to Biology

Complete Study Notes – Grade 9



Figure 1: DNA double helix – the blueprint of life.

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1.1 Definition of Biology

Biology is the scientific study of life or living things. The word comes from two Greek words: *bios* (life) and *logos* (study).

What does it mean to be 'alive'?

It is difficult to define life in a single sentence. However, we recognise life through common characteristics shared by all living systems:

- **Cellular organisation** – Living things are composed of one or more cells. Unicellular organisms (like bacteria) consist of a single cell, while multicellular organisms (like plants and animals) consist of many cells.
- **Growth** – Living things grow by increasing cell number or cell size. A tiny seedling grows into a large tree, and a baby grows into an adult.
- **Reproduction** – Living things can reproduce to create offspring, passing genetic information to the next generation. This ensures the continuation of the species.
- **Energy utilisation** – All living things need energy to perform work. Plants capture energy from sunlight through photosynthesis, while animals obtain energy by consuming food.
- **Homeostasis** – Living organisms maintain relatively constant internal conditions despite external changes. Humans maintain a stable body temperature around 37°C even when the environment changes.
- **Evolutionary adaptation** – Populations of living organisms change over time through natural selection, becoming better suited to their environments.
- **Response to stimuli** – Living things detect and respond to changes in their environment. Plants grow toward light, and animals flee from predators.

What is a Scientific Study?

Biologists study living things using the **scientific method**, which involves: asking questions about the natural world, suggesting possible answers (hypotheses), and testing these answers through observation and experimentation.

Relation to Other Subjects

Biology merges with other natural sciences. **Chemistry** explains molecular interactions and biochemical reactions in living organisms. **Physics** helps us understand energy flow, such as how plants convert radiant energy into organic molecules through photosynthesis.

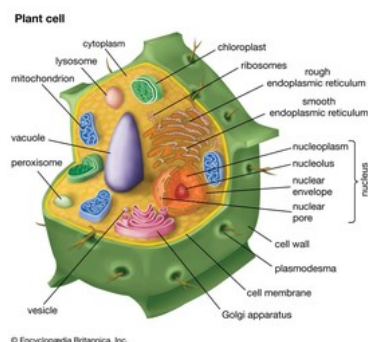


Figure 2: Plant cell structure showing major organelles.

1.2 Why Do We Study Biology?

Human curiosity – Humans have an inborn curiosity about the natural world, leading us to study living organisms and their habitats.

Understanding Life – Studying biology helps us understand how living things function and interact with their environment.

Applications in Daily Life:

- **Medicine and Healthcare** – Biologists discover drugs to treat diseases like AIDS, tuberculosis, and cancer. Understanding cellular and molecular mechanisms leads to better treatments and vaccines. Antibiotics like penicillin are produced by mould fungi or bacteria.
- **Forensics** – DNA analysis is used to investigate crimes. Biological techniques help identify victims and solve criminal cases.
- **Environmental Science** – Ecology helps societies evaluate environmental issues like global warming. Biologists study ecosystems and work to preserve biodiversity.
- **Food and Beverage Production – Fermentation** by yeasts, fungi, and bacteria is used in wine-making, brewing beer, baking bread, and producing cheese. Industrial chemicals like citric acid and lactic acid are produced using bacteria or fungi.
- **Waste Management** – Sewage disposal depends on bacteria in filter beds to purify effluent. Bacteria break down organic waste, protecting our water supplies.

Example: The discovery of the DNA structure by Watson and Crick in 1953 revolutionised biology and led to advances in genetics, medicine, and forensics.

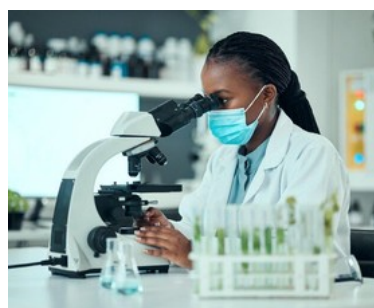


Figure 3: Biologist conducting research in a laboratory.

1.3 The Scientific Method

Biology is a **science of inquiry**. Biologists are always curious about why things happen or how things work.

The general steps of the scientific method are:

1. **Observation** – The scientific study begins with careful observations (often identifying a problem to solve). Observations can be made directly (using your sense organs) or indirectly (using scientific tools like microscopes).
2. **Asking Questions** – Observations lead the scientist to ask questions (inquiry). Example: 'Why do leaves change colour in autumn?'
3. **Forming a Hypothesis** – A **hypothesis** is a proposed scientific explanation (possible answer) for a set of questions. Several hypotheses can be proposed to solve a problem. Scientific hypotheses must be **testable**. Example: 'Leaves change colour because chlorophyll breaks down when temperatures drop.'
4. **Testing the Hypothesis** – Hypotheses are tested through **experimentation**. Scientific experiments must be **reproducible** – other scientists must be able to perform the same

experiment and achieve similar results. If a hypothesis is not supported by experimental data, a new hypothesis is proposed.

5. **Making Conclusions** – Scientists consider their original hypotheses in light of new information. If supported, the hypothesis remains a possible explanation. If not supported, the hypothesis is rejected, and alternative explanations are sought.

6. **Communicating Findings** – Scientists write papers explaining their methods and results. Papers are submitted to scientific journals in their field. Findings are shared with teachers, students, and the scientific community.

Real-World Example – Alexander Fleming and Penicillin

In the 1920s, Alexander Fleming observed that a mould (*Penicillium*) killed bacteria in a Petri dish. He questioned why this happened, hypothesised that the mould produced a substance that killed bacteria, and experimented by isolating that substance. This led to the discovery of **penicillin**, the first antibiotic.

1.4 Tools of a Biologist

Biologists use different types of tools in the laboratory and field for scientific investigations.

1.4.1 Laboratory Tools

Hand Lens	A convex lens fixed within a circular metallic loop attached to a wooden or metal handle. Provides an enlarged image of an object but is insufficient to observe details in cells. Used for examining insects, plant parts, or rocks.
Microscope	The most important tool for a biologist. An instrument used to study objects too small to be seen with the naked eye. Magnifies the image of small objects, making them visible to the human eye. Used to observe bacteria, fungi, parasites, and cell structures. Types: Light Microscope (uses beam of light) and Electron Microscope (uses beam of electrons, higher magnification). Key Terms: Magnification – number of times larger an image is; Resolution – ability to distinguish between two separate points.
Autoclave	Equipment used to sterilise (kill microorganisms) different biological samples. Exposes contaminated materials to high temperatures and highly pressurised steam. Kills microorganisms and their spores (which are resistant to chemical disinfectants).
Incubator	A device used to maintain a specific environment for culturing microorganisms. Maintains the temperature best suited for the growth of different types of microorganisms.
Petri Dishes	Flat dishes with a matching cover, available in glass or plastic. Used with gelling culturing media (like agar) to culture microorganisms. Placed in the autoclave for sterilisation.
Culture Tubes	Available in a range of sizes, shapes, and materials. Used to culture microorganisms in solid (gel) or liquid (broth) media. Also placed in the autoclave for sterilisation.

Flasks	Apparatus with a flat bottom and long narrow neck. Allows easy mixing of solution without spilling. Used to gently heat contents with a swirling motion. Essentially a large-volume culture tube used with broth media.
Balance	Useful for making up solutions accurately and weighing specimens.
Dropper (Pipette)	A glass tube with a small opening at one end attached to a rubber bulb. Used when it is required to control the amount of solution being added.
Tongs	Metallic scissors-shaped laboratory instruments. Used for lifting or picking up hot objects (heated crucibles, beakers, dishes, or flasks).
Dissecting Tool Kit	Used to dissect animals (frogs, fetal pigs, mice, etc.). Includes: scalpel, surgical scissors (straight and curved), forceps (with and without teeth), dissecting pins, and more.
Dissecting Pan	A pan on which the specimen is kept while being dissected to study its internal organs.
Crucible	A small container made of ceramic or metal. Able to withstand high temperatures. Used to melt elements.
Beaker	A cylindrical glass container. Used for making up solutions, holding hot or cold water, or ice baths.
Hotplate	Useful for heating solutions and making agar culturing media.
pH Meter	Used to measure the pH (acidity or basicity) of substances.
Thermometer	An instrument that measures the temperature of substances.
Forceps	Used to hold or pick up small objects. Available in various shapes and sizes.
Spatula	Shaped like a spoon. Used for mixing substances into a solution, stirring, and scraping objects.
Wash Bottles	Used to rinse various laboratory materials. Flexible, allowing users to adjust water pressure by squeezing the bottle.
Bunsen Burner	Produces an open flame. Used for sterilising inoculation loops and heating.



Figure 4: Petri dishes used to culture microorganisms on agar.

1.4.2 Field Tools

Insect Net – Composed of a net bag made of cloth or fine mesh attached to a wire loop. The wire loop is attached to a wooden or metal pole. Used to catch flying or hopping insects.

Fishing Net – A net used for fishing. Devices made from fibres woven in a grid-like structure. Some fishing nets are also called fish traps.

1.5 The Light Microscope

Most microscopes have several different powerful lenses attached to them, allowing the viewer to inspect content at more than 100 times its actual size.

Importance: Biology as modern science would not exist without the microscope. Good microscopes are essential for most biologists. Microscopes are extremely expensive, so it's important to know the function of all structures and handle the device properly.

How it Works:

Light rays are focused onto the image on a microscope slide. Light transmitted through the specimen is focused by two types of lenses: **eyepiece** and **objective** lenses. A compound light microscope magnification is the **product** of eyepiece and objective lenses. Example: $10\times$ eyepiece $\times 40\times$ objective = $400\times$ total magnification.

1.5.1 Parts and Functions of the Light Microscope

Eyepiece (Ocular)	Part found at the top of the microscope. Used to look through the microscope. Standard magnification is $10\times$ (optional $5\times$ to $30\times$).
Eyepiece Tube	The eyepiece holder. Carries the eyepiece just above the objective lens. In binocular microscopes, the eyepiece tube is flexible and can be rotated.
Objective Lenses	Major lenses that further magnify the specimen. Multiple objective lenses with different magnifications ($4\times$, $10\times$, $40\times$, $100\times$).
Arm	A holder connected to all components. Functions as a support for the microscope. Used for carrying the microscope.
Body Tube	Connects the eyepiece to the objective lenses. Maintains the optical path alignment.
Nose Piece (Revolving Turret)	Rotating mount that holds multiple objective lenses. Movable, allowing the user to change magnification.
Coarse Adjustment Knob	Brings specimen into general focus. Moves the stage up and down quickly.
Fine Adjustment Knob	Regulates the distance between object and objective for sharpness. Moves the stage only minimally.
Stage	The section where the specimen is placed for viewing. Flat platform with clips to hold slides.
Stage Clip	Acts as a holder for the object plate. Ensures the slide doesn't fall out of place.
Aperture	A hole in the microscope stage. Allows transmitted light from the source to reach the stage.
Microscopic Illuminator	The microscope's light source located at the base. Used instead of a mirror. Captures light from an external source (low voltage, about 100V).

Condensers	Lenses used to collect and focus light from the illuminator into the specimen. Found under the stage next to the diaphragm. Essential for producing clear, sharp images at high magnification (400× and above).
Diaphragm (Iris)	Found under the stage. Primary role is to control the amount of light reaching the specimen. Adjustable apparatus controlling light intensity and beam size.
Base	The very bottom part. Serves to accommodate all parts of the light microscope. Provides stability and support.

Calculating Total Magnification: Multiply the eyepiece power by the objective power. For example, a 10× eyepiece with a 40× objective gives 400× magnification.

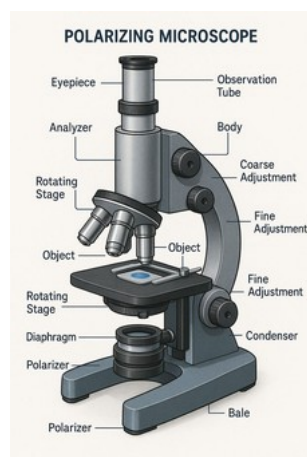


Figure 5: A compound light microscope – essential for viewing cells.

1.5.2 Handling and Using a Light Microscope

- **Proper Handling:** Be very careful when removing the microscope from the cabinet. Carry the microscope properly – grip the microscope by the **arm** and put your hand beneath the **base**. Hold the microscope upright at all times.
- **Using the Microscope:** Ensure the slide is clean and dry before putting it on the stage. Place the slide on the stage with the most promising region exactly in the middle of the aperture.
- **Focusing:** Always focus on **low power (4× objective) first** – even if you eventually need high power magnification. Focus with the **larger coarse-focusing knobs** first. When you have nearly got the image in focus, make it really sharp using the **smaller fine-focusing knobs**.
- **Increasing Magnification:** Move the slide so the most promising region is exactly in the middle of the field of view, then change to a higher magnification lens.
- **Oil Immersion:** Use **immersion oil only with the 100× objective** (oil immersion lens). Use only **one drop** of oil.
- **Focusing Direction:** Always focus by moving the lens and the specimen further apart – never closer together.
- **Cleaning:** Never touch the surfaces of the lenses with your fingers or anything else. Always clean the microscope when you are done (use lens paper and alcohol).
- **Storage:** Lower the stage and remove the slide when you are done. Always place the 4× objective over the stage and ensure the stage is at its lowest position before putting the microscope away.

1.6 General Laboratory Safety Rules

Safety is the condition of being protected from harm or other danger. Laboratory safety is important before we do any activities in the laboratory.

Important Laboratory Safety Rules:

- **Following Instructions** – Read and listen to the laboratory procedure. Be familiar with all steps from start to finish. Know how to use all lab equipment before you begin.
- **Knowing the Location of Safety Equipment** – Locate the **fire extinguisher** and **first-aid kit**. Know the location of safety equipment and how to use it. Periodically check equipment to ensure it is in working order. Review lab safety signs before starting an experiment.
- **Dressing for the Laboratory** – Wear a **laboratory coat, safety goggles, gloves**, and appropriate clothing. Wear long pants and closed-toe leather shoes (NO sandals). Tie back long hair and avoid loose clothing. Wear a disposable respirator mask when handling toxic chemicals.
- **Never Eat or Drink in the Laboratory** – No food or drink in the science laboratory. Do not store food or beverages in the same refrigerator as chemicals or cultures.
- **Never Taste or Sniff Chemicals** – Avoid tasting or smelling chemicals or biological cultures. Label containers and read before use. Get in the habit of making a label for glassware before adding chemicals.
- **Act Responsibly in the Laboratory** – Never randomly mix chemicals to see what happens. This may result in explosion, fire, or release of toxic gases.
- **Cleaning the Experiment Area** – Every laboratory session should begin and end with clean, properly stored glassware, chemicals, and equipment.
- **Handling Chemicals Properly** – Wear a disposable respirator mask when handling toxic chemicals. Never allow laboratory chemicals to contact your bare skin.
- **Knowing What to Do with Laboratory Accidents** – If someone is burned or exposed to chemicals, immediately flood the burned area with cold tap water for several minutes. This minimizes damage from the burn or chemical exposure.
- **Place Emphasis on Safety at All Times** – Never leave containers of chemicals open. All containers must have appropriate labels. Unlabelled chemicals should never be used.

End of Unit 1: Introduction to Biology

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